

Quiz 3 — Testing a Coin

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices · same skills as Practice 4

Name: _____ UT EID: _____

1. Each entry of $\mathbf{X}$ is an independent $N(0, 1)$ sample. Predict the first output exactly; for the other two, give the values they approach as the number of rows grows. Do not run the code. [5 pts]

```
import numpy as np
rng = np.random.default_rng(42)
X = rng.normal(size=(10_000, 100))
q = (X > 0).mean(axis=1)
print(q.shape)
print(q.mean())
print(q.std())
```

Answer:

2. In 100 coin flips, 60 are heads. The code uses the approximation $D(\hat{q}||p) \approx 2\varepsilon^2$, where $\varepsilon = k/n - 1/2$ and $p = (1/2, 1/2)$. The test rejects when $2nD > 3.84$. Write all four outputs. [5 pts]

```
n, k = 100, 60
eps = k/n - 0.5
D = 2 * eps**2
T = 2 * n * D
print(round(eps, 2))
print(round(D, 2))
print(round(T, 2))
print(T > 3.84)
```

Answer:

Bonus. If exactly 60% of n flips are heads, what is the smallest possible n for which the same approximate test rejects? Show the inequality you solve. (An exact 60% split requires n to be a multiple of five.) [2 bonus pts]

Answer:

Total: 10 points + 2 bonus points. Lowest two quiz scores are dropped at the end of the semester.